

Deep Site-Occupancy Models for Plant species temporal **Trend** assessment from **Citizen Science** Data

Raphaël Benerradi, PhD student at Inria-LIRMM (Montpellier, France)

Supervision: Alexis Joly, Christophe Botella, Maximilien Servajean

World Biodiversity Forum - Session IND11 - 17/06/2026



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Pl@ntNet

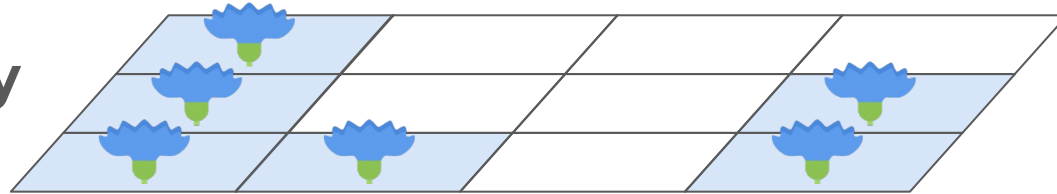


LIRMM

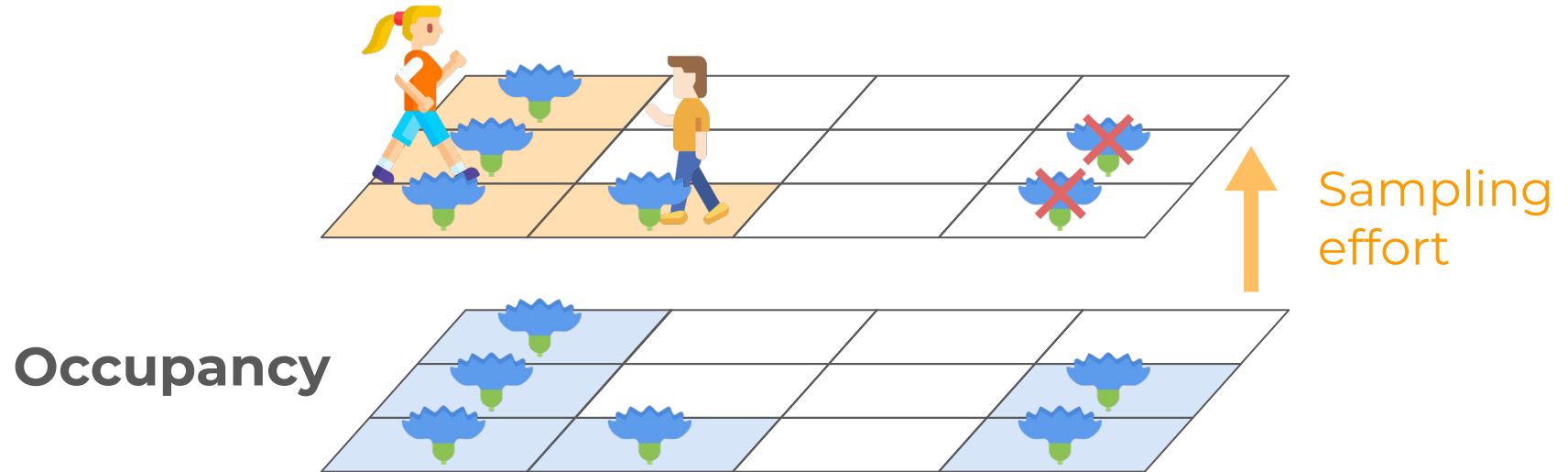
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Modeling occupancy

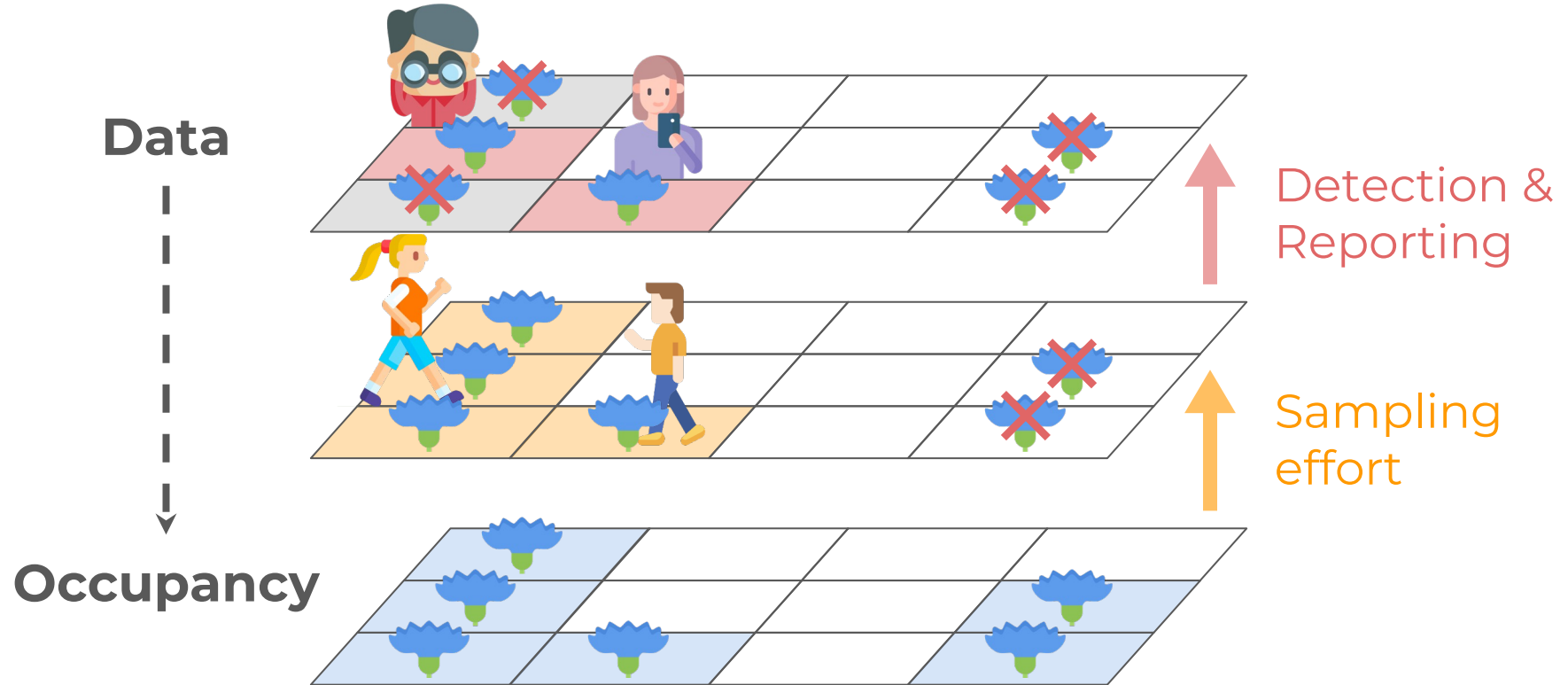
Occupancy



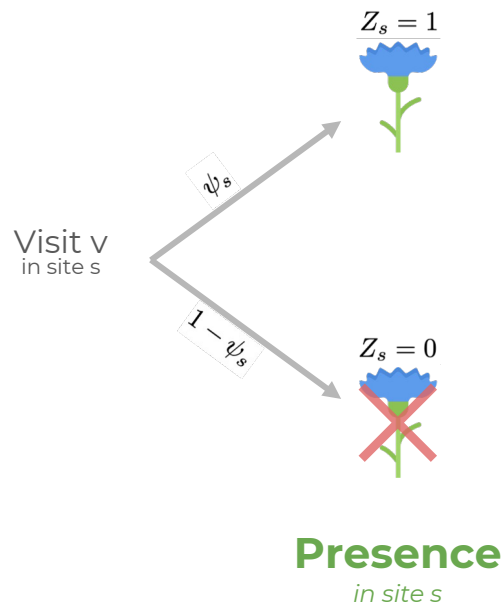
Modeling occupancy



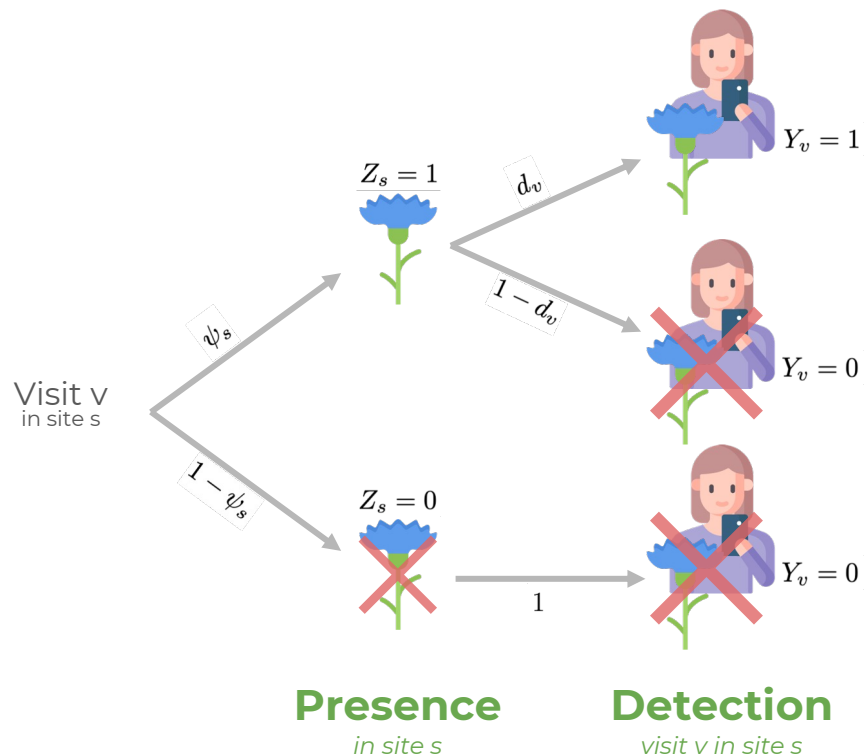
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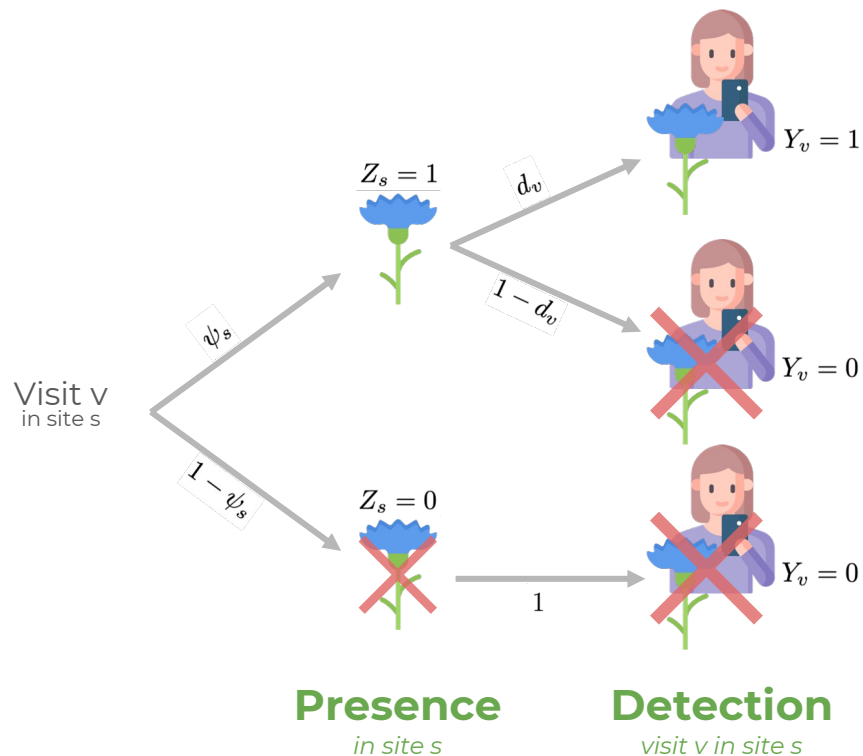
Single-season single-species



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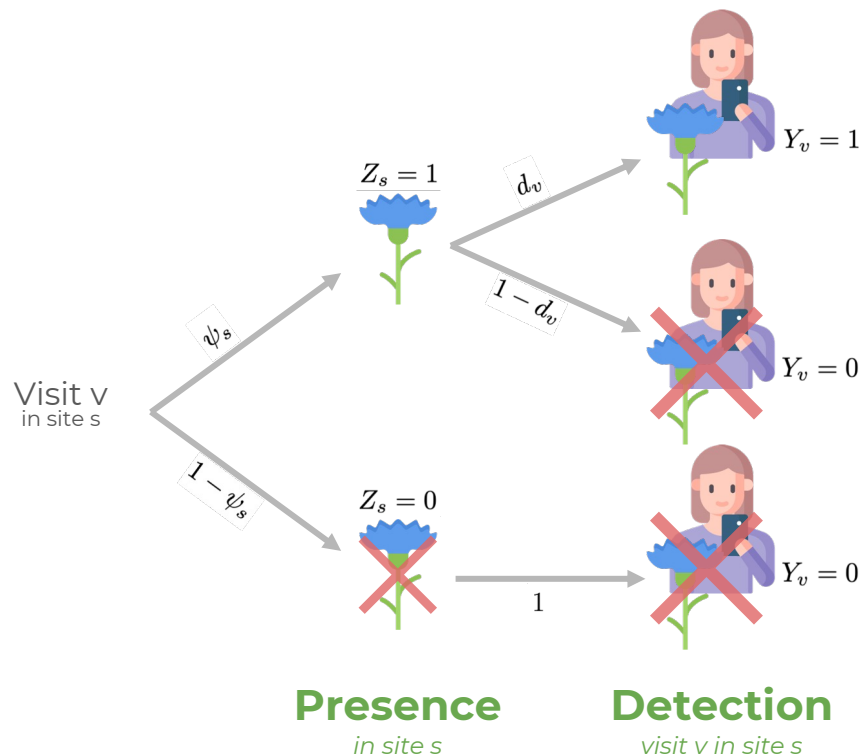


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$$\text{(Detection)} \quad Y_v | Z_{s(v)} \stackrel{\text{ind.}}{\sim} \mathcal{B}(d_v \cdot Z_{s(v)})$$

- **Explicit likelihood** to maximize
(only depending on ψ_s , d_v and y_v)

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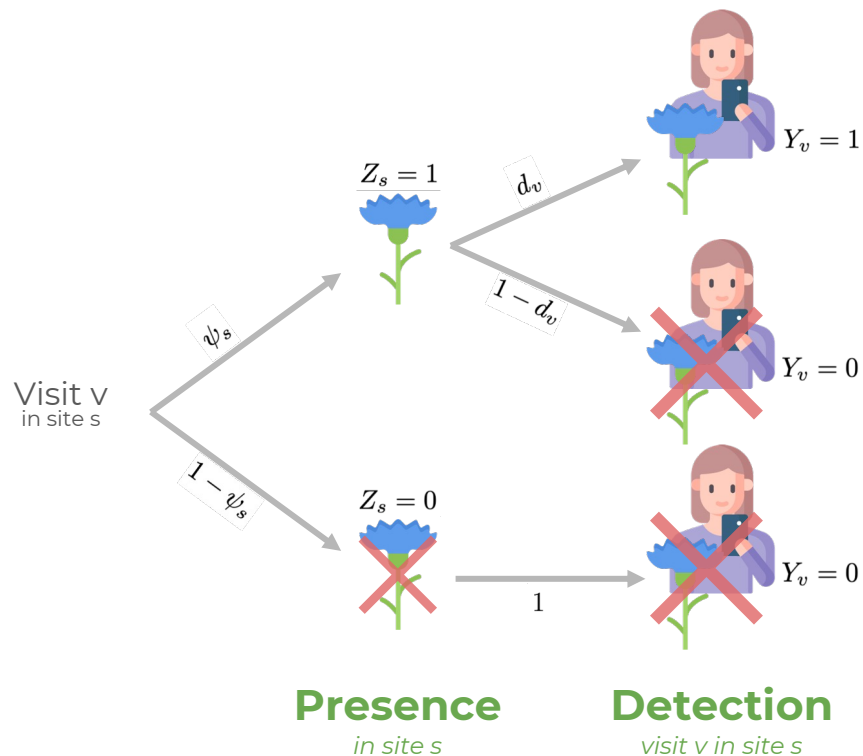


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 - Sigmoid of linear predictors
 - Sigmoid of **neural network**
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- **Multiple independent species:** product of the likelihood

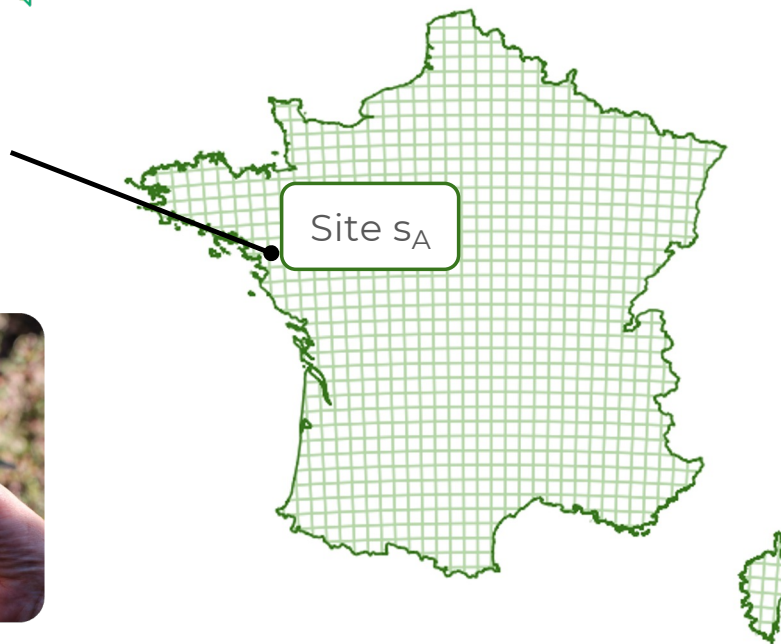
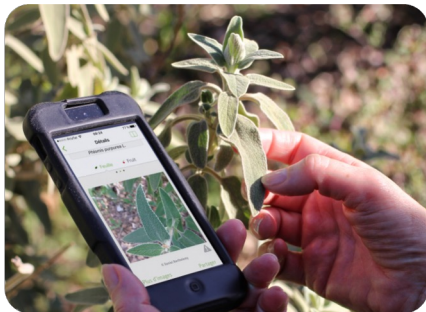
Citizen science programs - Opportunistic data

e.g.  Pl@ntNet

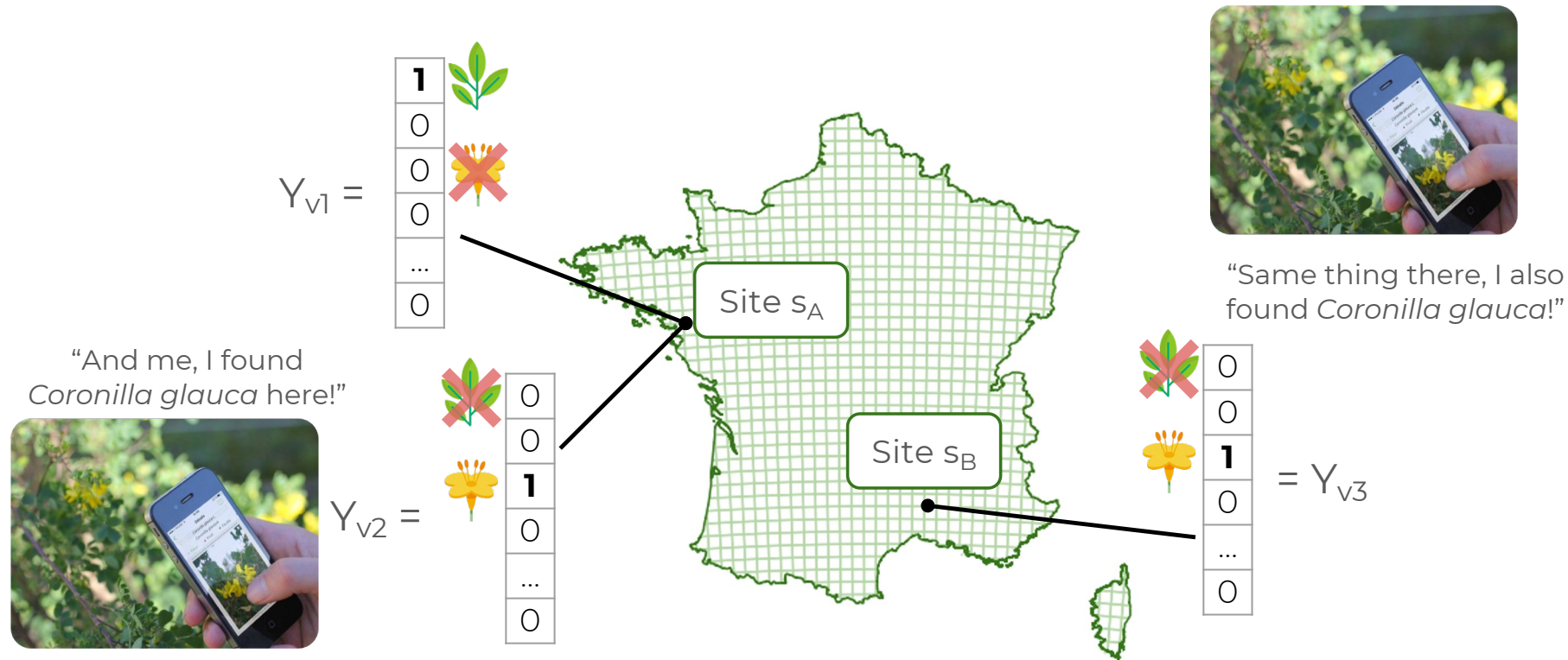
For K species

$$Y_{v1} = \begin{Bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \dots \\ 0 \end{Bmatrix}$$


"Hey, I found *Phlomis purpurea* here!"



Citizen science programs - Opportunistic data



Challenges with these data

Massive datasets (e.g. a subset of **Pl@ntNet** in metropolitan France):

- **2,605,648 observations**
- **12,289 species**

Challenges with these data

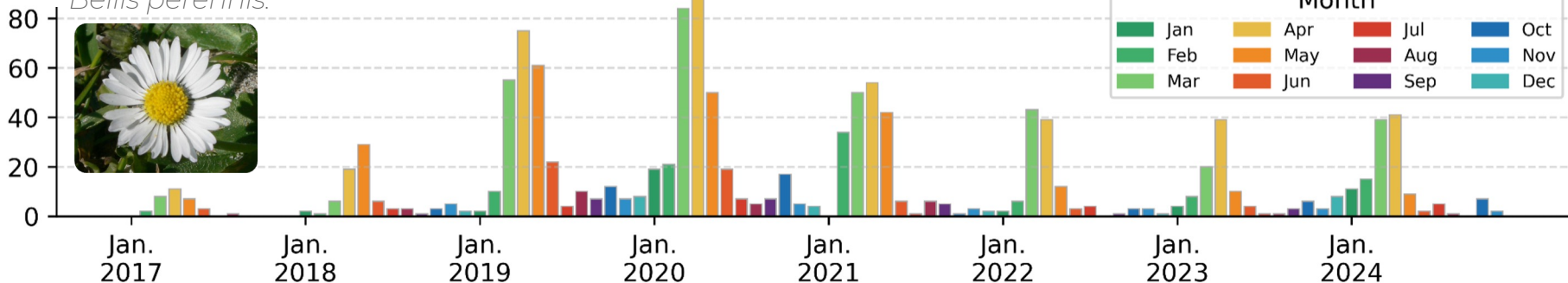
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Unbalanced over space and time:

- Paris: 971.7 observations per km²
- Creuse: 1.1 observations per km²

Number of detections per month
Bellis perennis:



Challenges with these data

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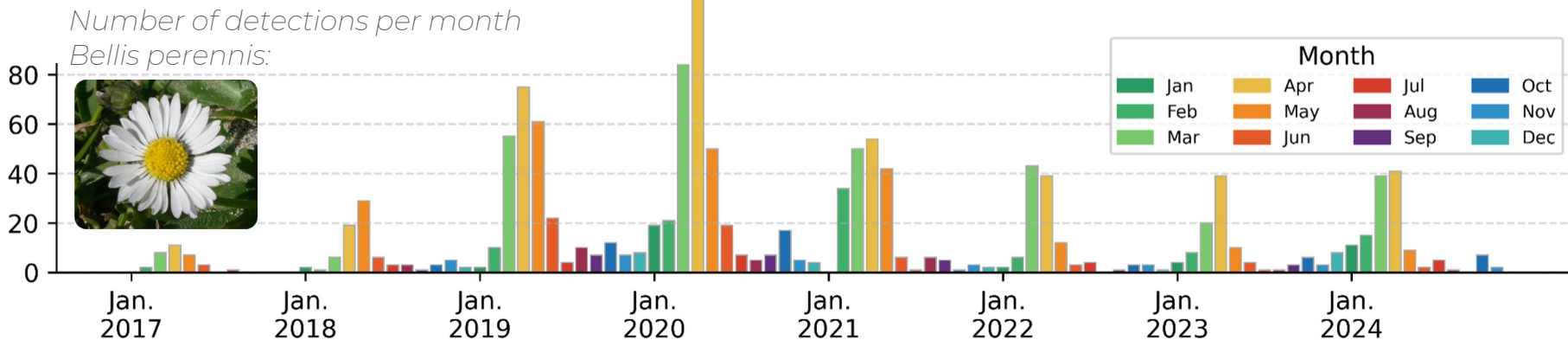
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→ Require **efficient** implementation
(data structure, optimizer...)

→ **Heterogeneous and complex** patterns
(non-linear effects, detection process...)



OccuPYcy *(in prep)*

- Implemented in  python™, relying on  PyTorch, and inspired by the syntax of `unmarked` (R package)

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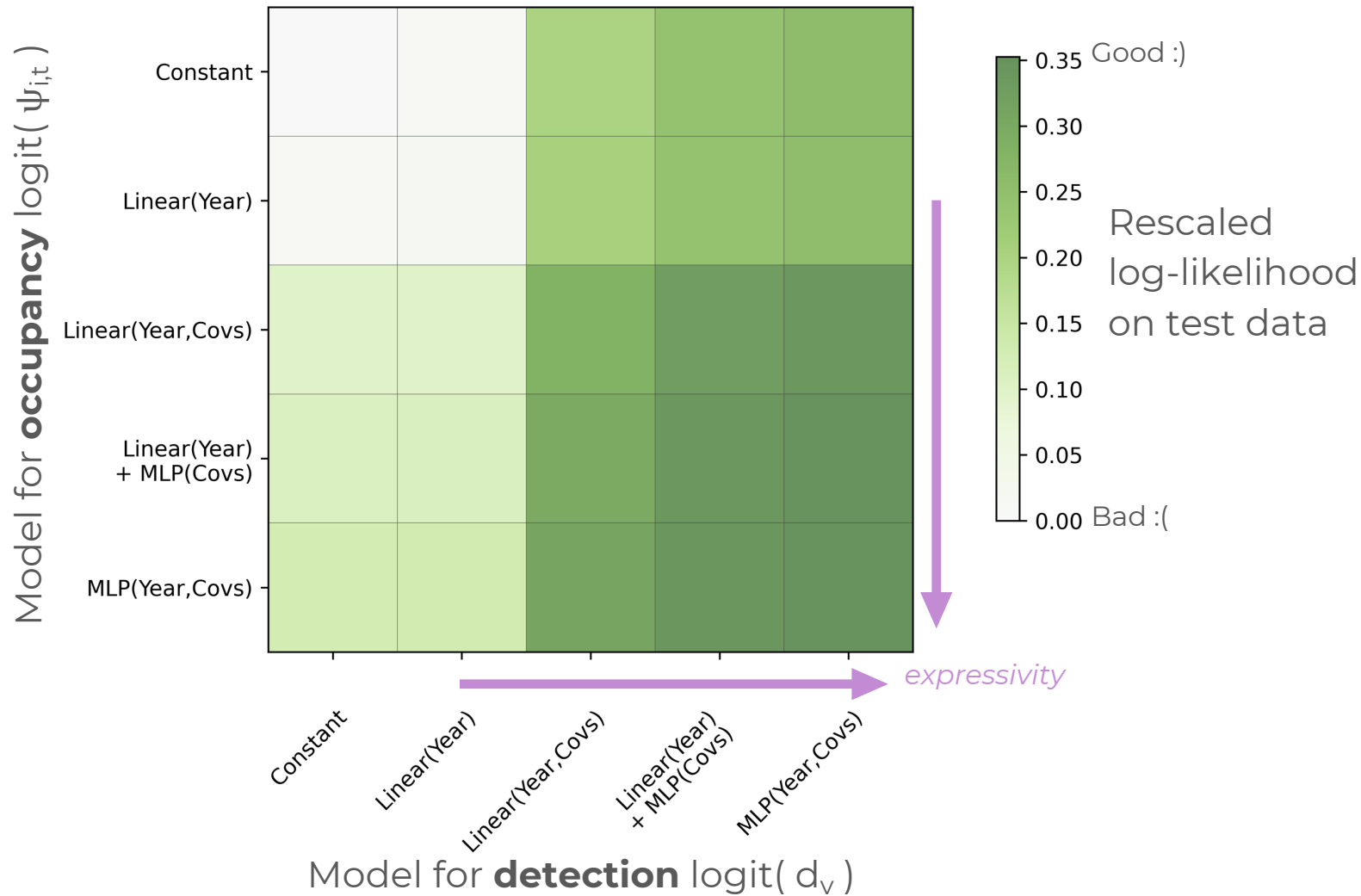
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- **Maximum Likelihood Estimation** – multiple optimization methods
 - Traditional optimizers: LBFGS
 - Stochastic optimizers: SGD, Adam...
 - Regularizations
 - Hyperparameter tuning

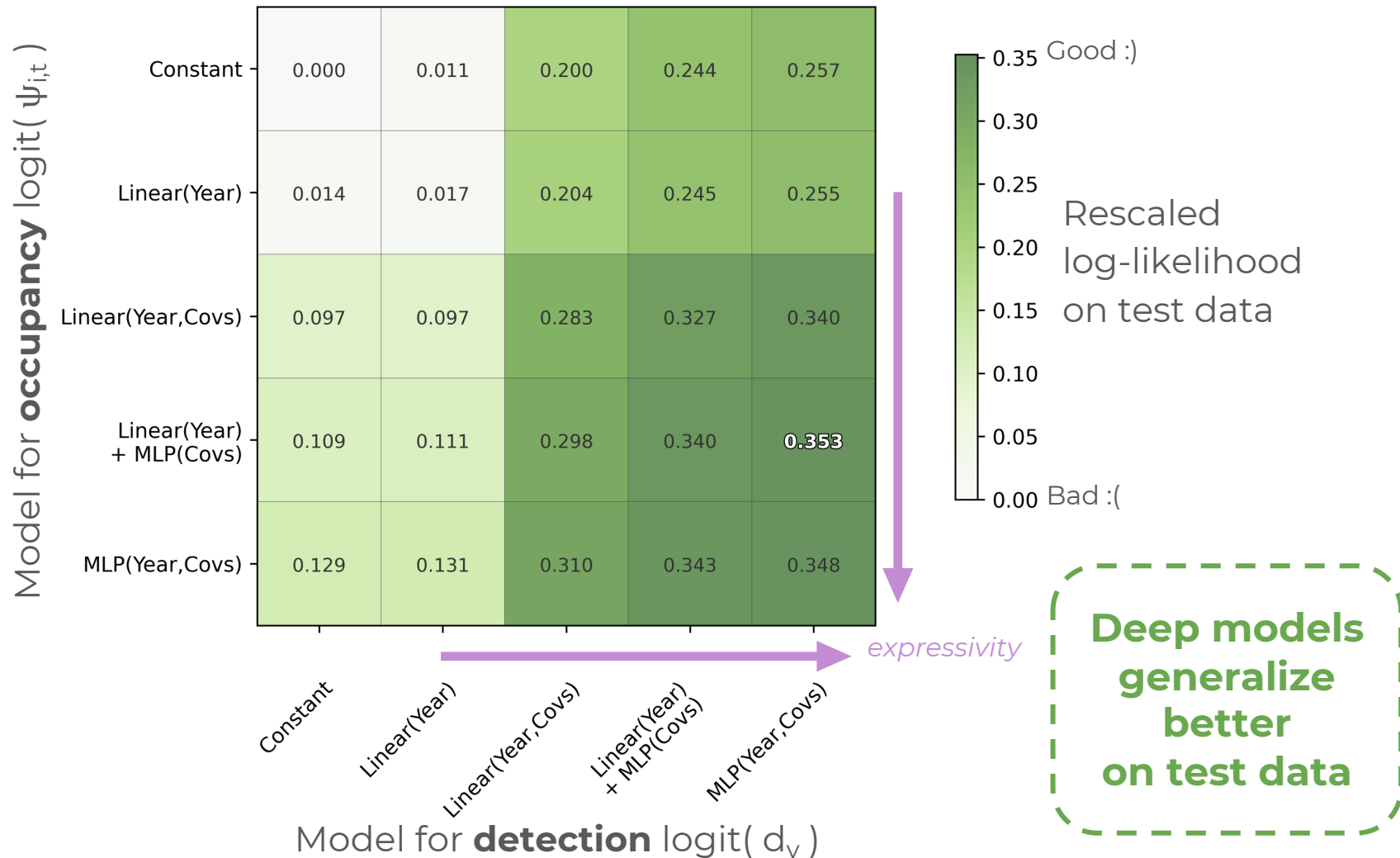
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 - Ready to use models
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- **Flexible models** for occupancy and detection
 - Ready to use models
 - Customizable options
- Suitable for **large, heterogeneous datasets**

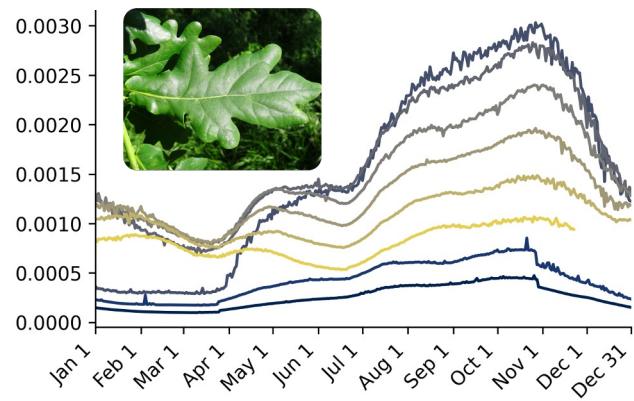




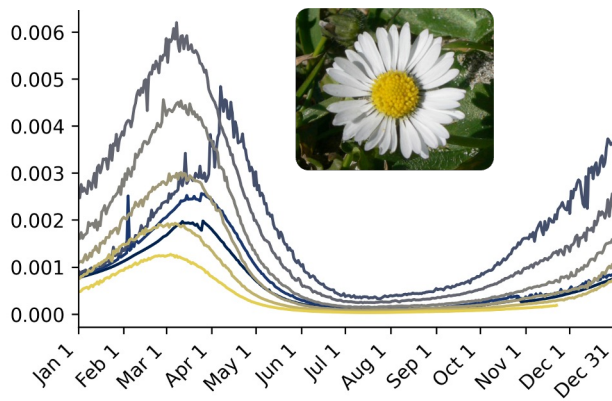
Fitted detection probability



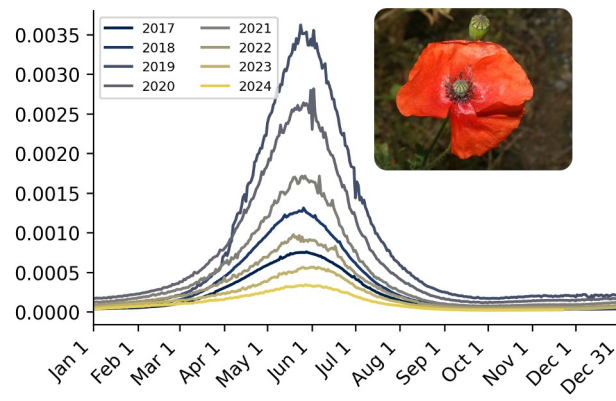
Quercus robur



Bellis perennis



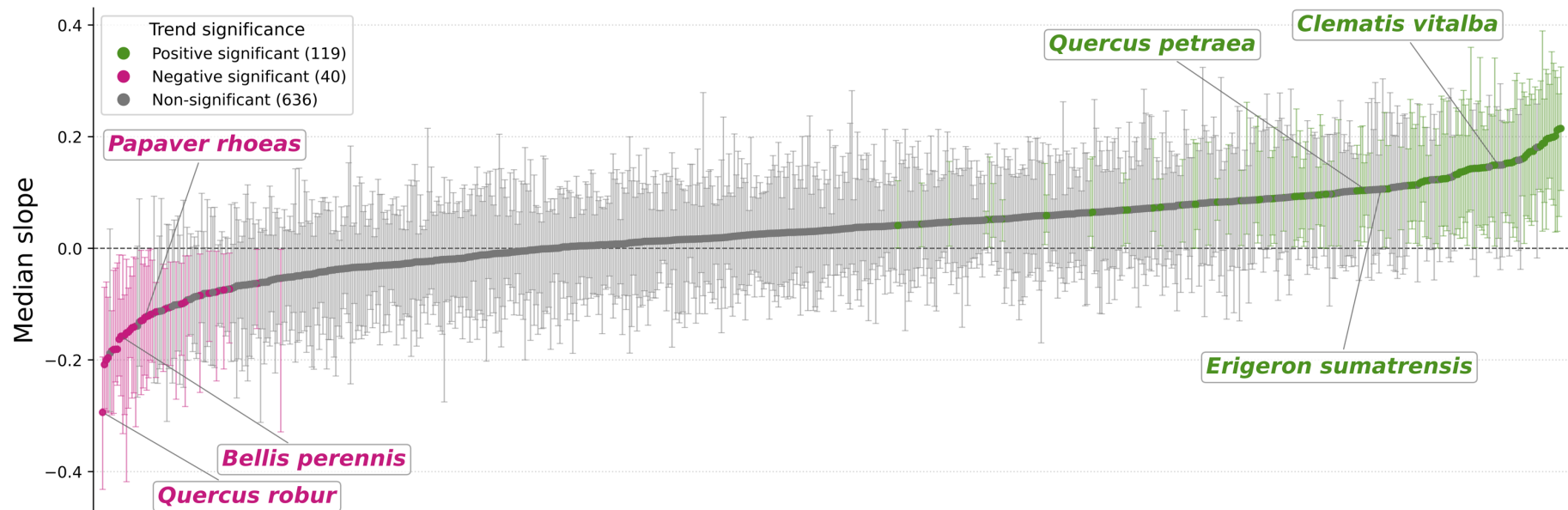
Papaver rhoeas



Occupancy: $\text{logit}(\psi_{i,t}) = \text{coef} \times \text{year} + \text{MLP}(\text{bioclim}, \text{land use})$

Detection: $\text{logit}(\mathbf{d}_v) = \text{MLP}(\text{year}, \sin\text{-cos Day of year}, \sin\text{-cos Time in day})$

Trend estimation



Over 795 species based on 2,605,648
PI@ntNet observations
in metropolitan France (sites = yearly 25 x 25 km² cells)

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Key messages

- **Occupancy models** can be applied on **opportunistic citizen science** data to take advantage of their large volume
- They enable **bias correction** for species **trend** estimation
(preprint coming soon)
- **Deep learning** architectures can be incorporated into occupancy models (OccuPYcy package coming soon)

thanks for your attention

Supervisors:



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Maximilien
Servajean,
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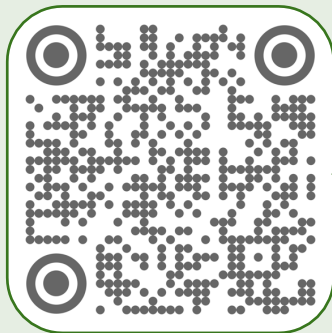
Christophe
Botella,
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Raphaël BENERRADI

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Github: <https://github.com/RaphBnrd/>



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Pl@ntNet



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Bibliography

1. MacKenzie, D. I., Nichols, J. D., Lachman, G. B., Droege, S., Andrew Royle, J., & Langtimm, C. A. (2002). Estimating Site Occupancy Rates When Detection Probabilities Are Less Than One. *Ecology*, 83(8), 2248-2255. [https://doi.org/10.1890/0012-9658\(2002\)083%255B2248:ESORWD%255D2.0.CO;2](https://doi.org/10.1890/0012-9658(2002)083%255B2248:ESORWD%255D2.0.CO;2)
2. Guillera-Aroita, G. (2017). Modelling of species distributions, range dynamics and communities under imperfect detection : Advances, challenges and opportunities. *Ecography*, 40(2), 281-295. <https://doi.org/10.1111/ecog.02445>
3. Joseph, M. B. (2020). Neural hierarchical models of ecological populations. *Ecology Letters*, 23(4), 734-747. <https://doi.org/10.1111/ele.13462>